Williams®

16P-3001-101
February, 1981

DEFENDER



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Williams®
ELECTRONICS, INC.

3401 N. California Ave., Chicago, IL 60618
Cable Address: WILCOIN, CHICAGO
(312) 267-2240

SPECIAL NOTES

CPU/Video Board

Note modification required on all sample games described in Service Bulletin V-1. Should failure of 4D 7486 IC occur, make foil cuts on board as described in Service Bulletin V-3.

ROM Board

When blue-labeled ROMs are used, ROM 4 is not used, jumpers W2 and W3 are connected, and jumper W1 and W4 are removed. When green-labeled ROMs are used, ROM 4 is used, jumpers W1 and W4 are connected, and jumpers W2 and W3 are removed. Between blue and green-labeled ROM sets, ROMs 2, 6, 7, 8, 9, 10, 11, and 12 are compatible.

Note foil cuts and addition of resistors required as described in Service Bulletin V-2.

Interface Board

Note foil cuts and addition of resistors required as described in Service Bulletin V-2.

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No. V-1

SERVICE BULLETIN

DEFENDER CPU/VIDEO BOARD
MODEL D8356

Watchdog Circuit Modification

INTRODUCTION
The Watchdog timer on D8356 CPU/Video Board in sample games has exhibited a tendency to timeout too quickly. This results in the game intermittently resetting. Should this occur with credits in the game, the credits will be removed.

MODIFICATION
All CPU/Video Boards in sample games shipped prior to 12-8-80 must be modified as follows. The modification consists of cutting a trace and adding a jumper. Refer to the drawing on the other side of the sheet and proceed as follows:

1. Locate the trace from 3C pin 13 on the solder side of the board. 3C is a type 74393 or 74LS393.

2. Cut the trace located in step 1.

3. Obtain a 6 inch length of 30-gauge or other availabe gauge of insulated wire.

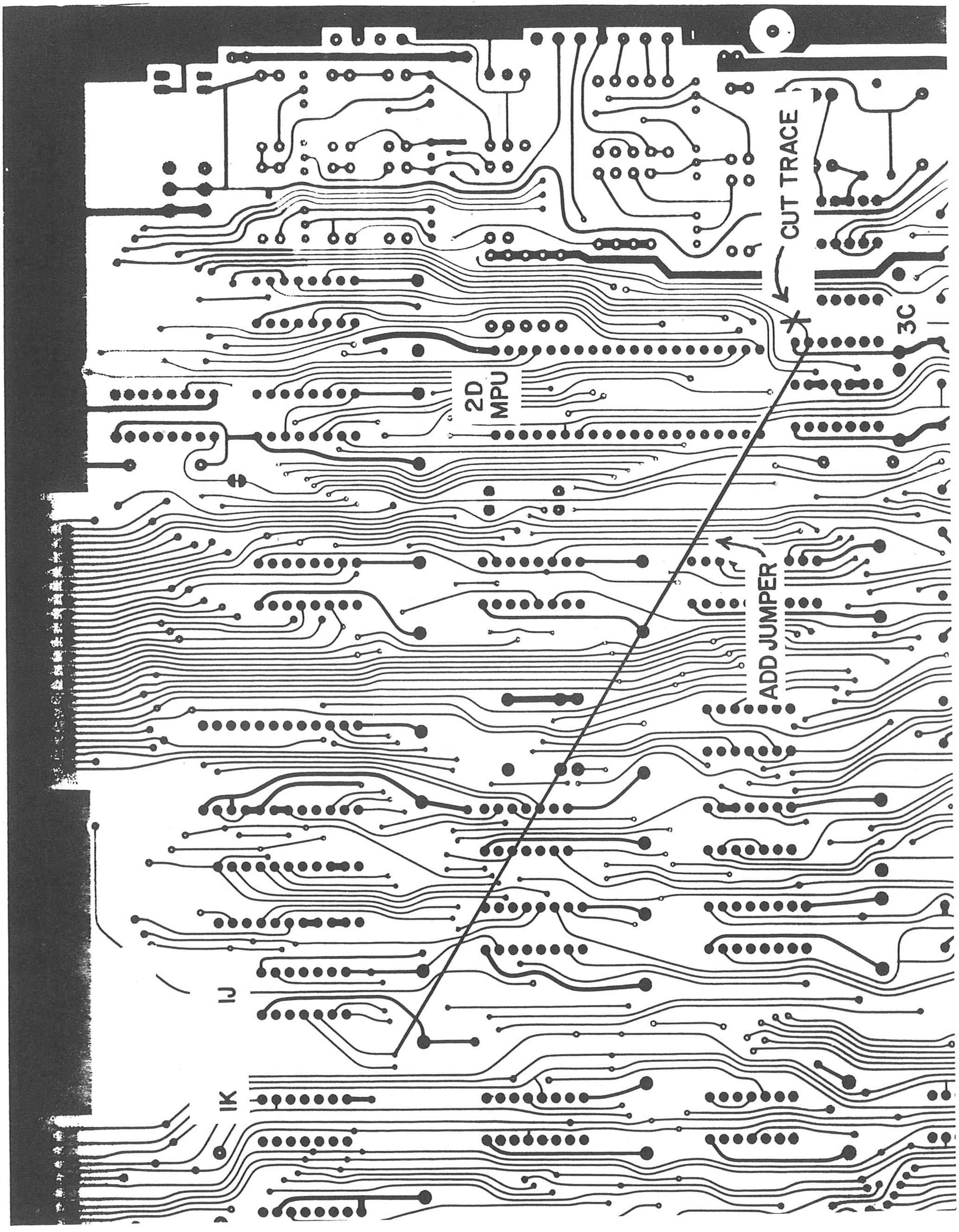
4. Strip back the insulation at each end of the wire approximately 1/8-inch and tin the wire.

5. Solder one end of the wire to 3C pin 13 on the solder side of the board.

6. Locate the trace connected to 1J pin 10. 15 is a type 7404.

7. Solder the other end of the wire to the feedthru connected to 1J pin 10.

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CUT TRACE
3C
2D
MPU
ADD JUMPER
J
IK

Williams®

No. V-2

SERVICE BULLETIN

DEFENDER ROM AND INTERFACE BOARDS
MODELS D8357 AND C8358
Addition of Resistors

INTRODUCTION
Switch inputs to the ROM Board and Interface Board are applied to CMOS hex inverters which are susceptible to possible damage from static discharge. Addition of series resistors is required to provide protection.

MODIFICATION
This modification must be made to these boards in all sample games. Refer to the drawings on the reverse side and proceed as follows:

1. Obtain 23, 1K ±10%, 1/4 W resistor (Williams Part No. 5A-8984)

2. Cut the traces as indicated in seven places on the ROM board.

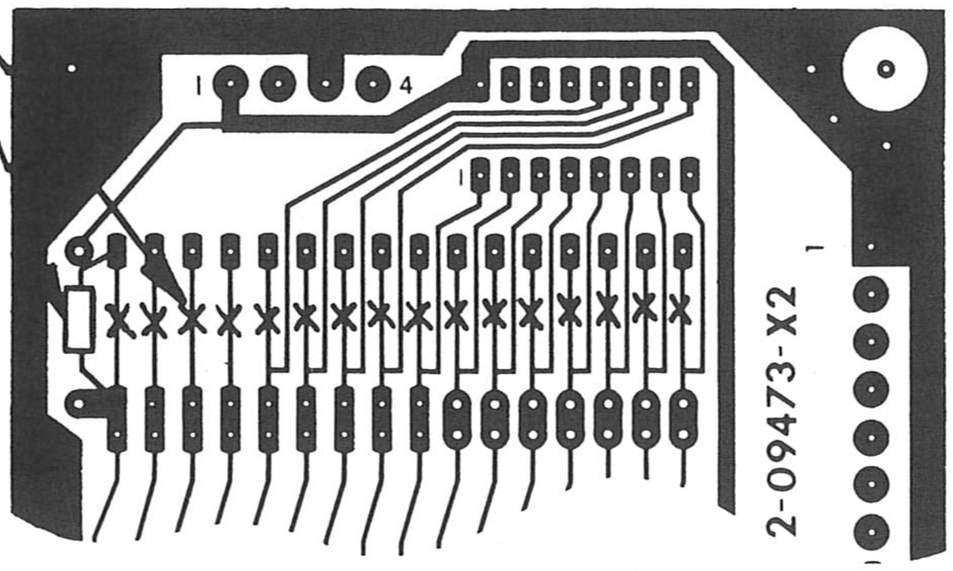
3. Solder in a resistor bridging each cut trace in the pads on the board.

4. Repeat steps 2 and 3 for the Interface board, cutting 16 traces.

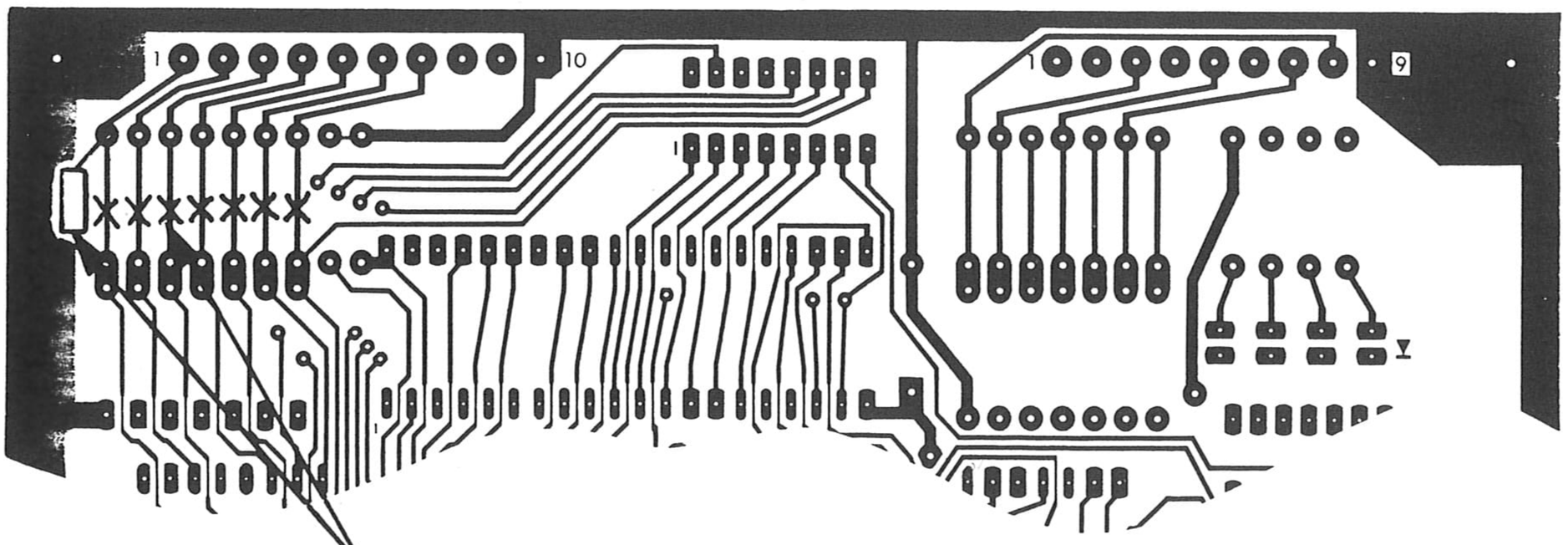
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C 8358
INTERFACE BOARD

CUT 16 TRACES,
ADD 16 RESISTORS



D 8357
ROM BOARD



CUT 7 TRACES,
ADD 7 RESISTORS

Williams®

No. V-3

SERVICE BULLETIN

DEFENDER CPU/VIDEO BOARD
MODEL D8356
Foil Cuts at 4D 7486 IC

INTRODUCTION

Termination of pins for unused gates in the 4D 7486 IC may cause overheating in ICs with certain pin configurations. This could cause the IC to fail.

MODIFICATION

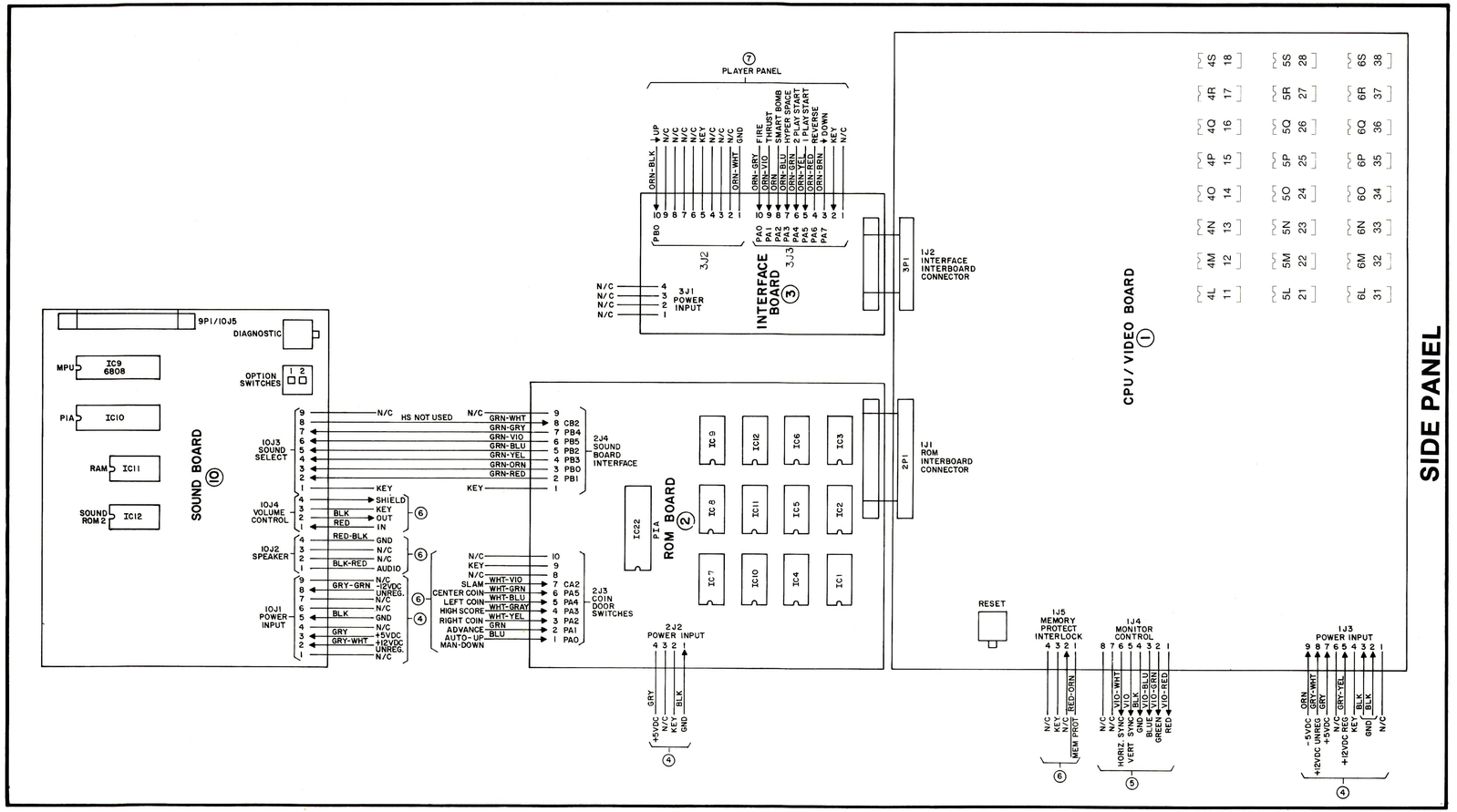
Failure will normally occur within a few days of operation so it is recommended not to make the modification unless replacement of the 4D 7486 IC is required. When replacing the chip, proceed as follows:

1. On the foil side of the board, remove solder from pins 6 and 7 and pins 8 and 9 of the 4D IC.

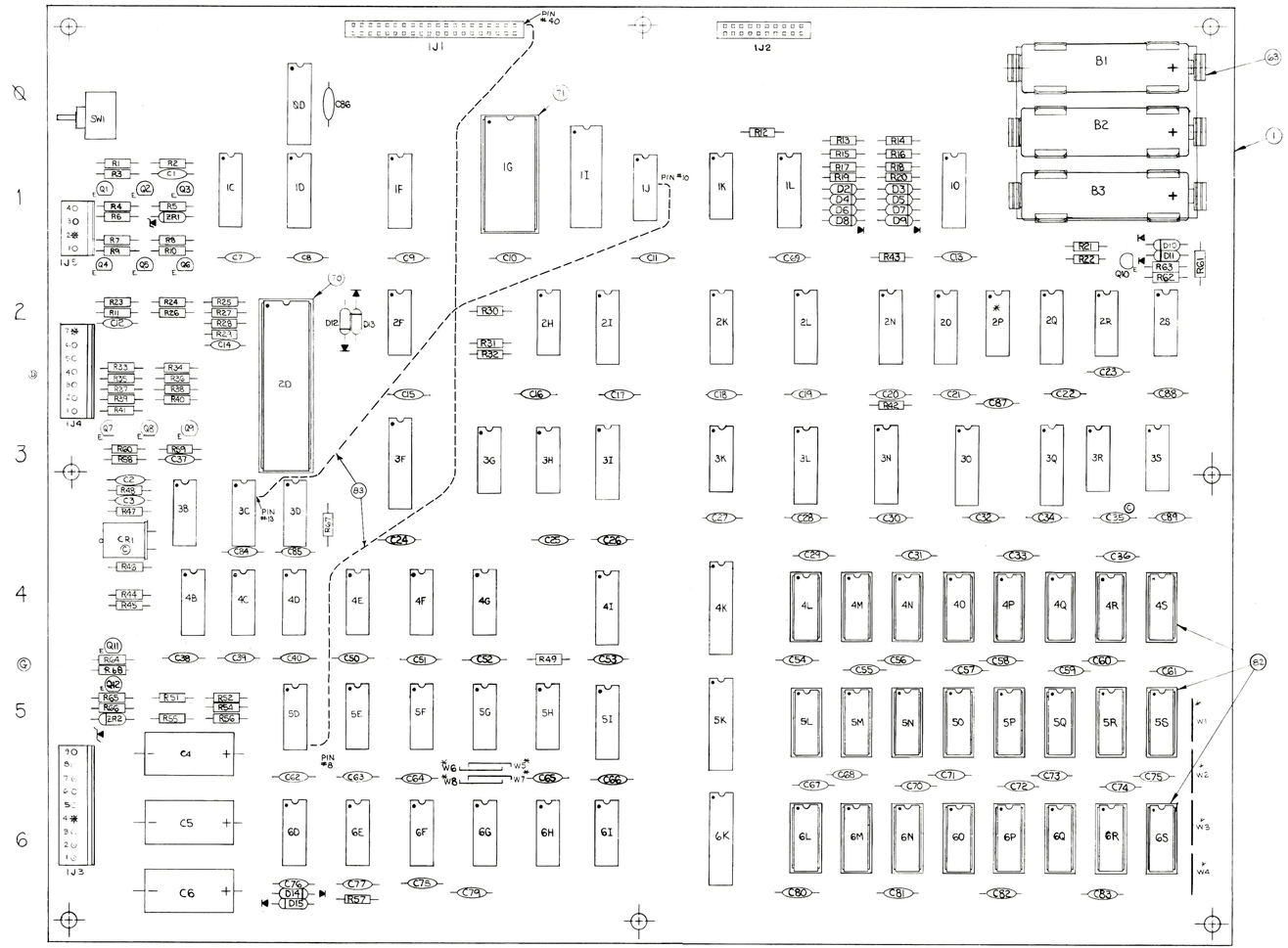
2. Cut the foil between pins 6 and 7 and between pins 8 and 9.

3. Carefully resolder pins 6, 7, 8, and 9, making sure that solder does not bridge across the pins.

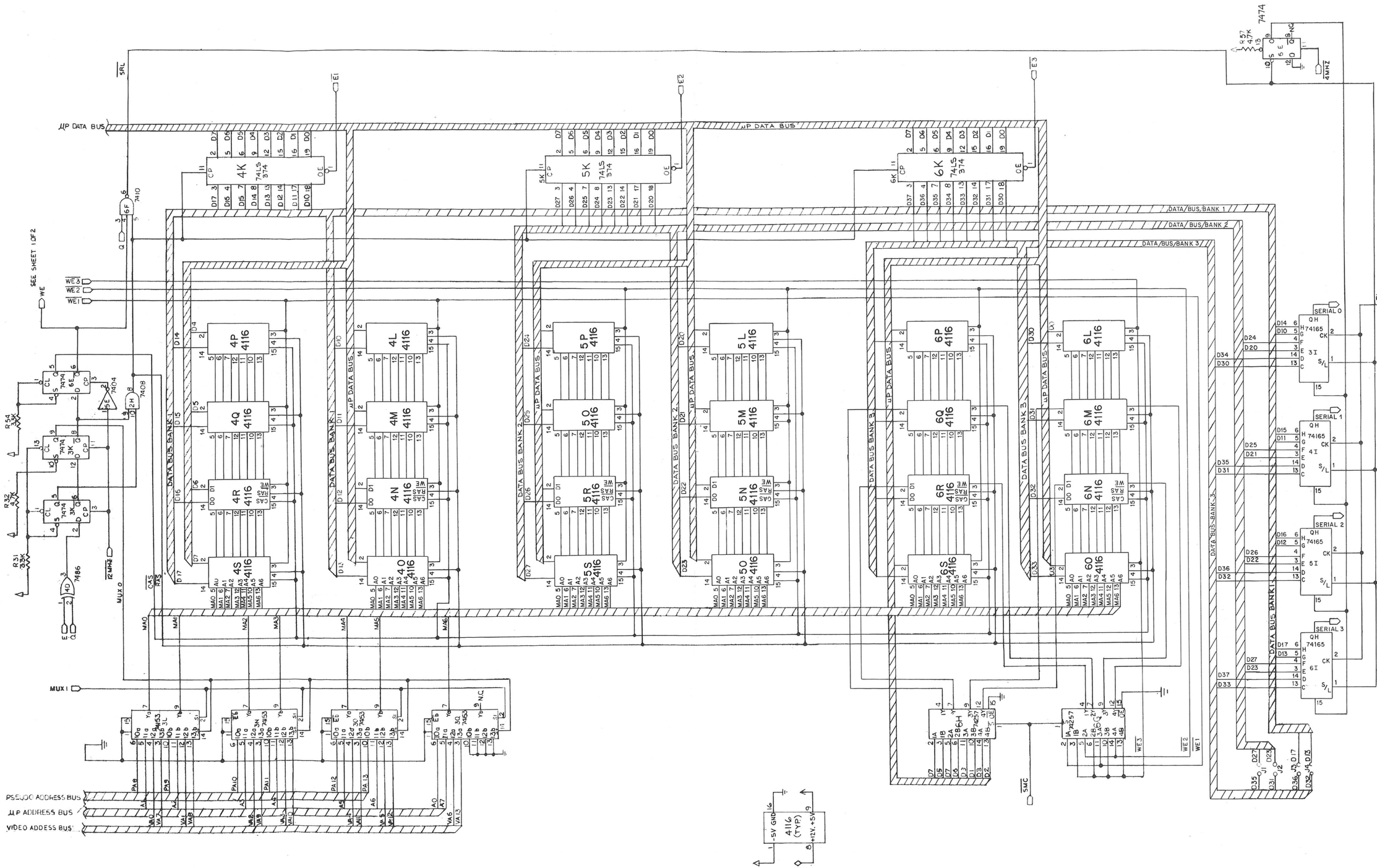
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Interboard Wiring Diagram


A B C D E F G H I J K L M N O P Q R S
1 2 3 4 5 6
1J1
1J2
1J3
1J4
1J5
SW1
B1
B2
B3
C1
C2
C3
C4
C5
C6
CR1
Q1
Q2
Q3
Q4
Q5
Q6
Q7
Q8
Q9
Q10
Q11
Q12
Q13
Q14
Q15
Q16
Q17
Q18
Q19
Q20
Q21
Q22
Q23
Q24
Q25
Q26
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Q35
Q36
Q37
Q38
Q39
Q40

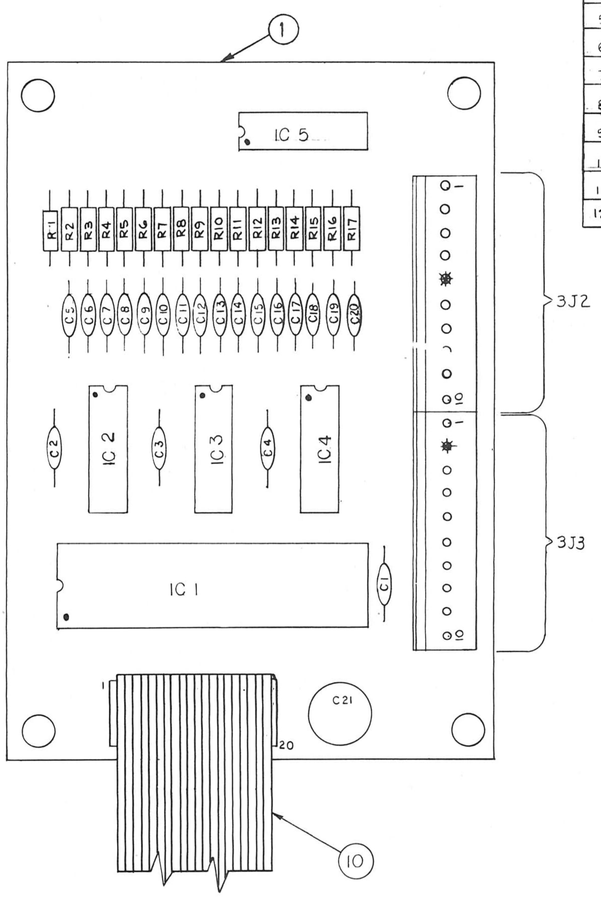
D8356 CPU/Video Board Logic Diagram
(Sheet 1 of 2)

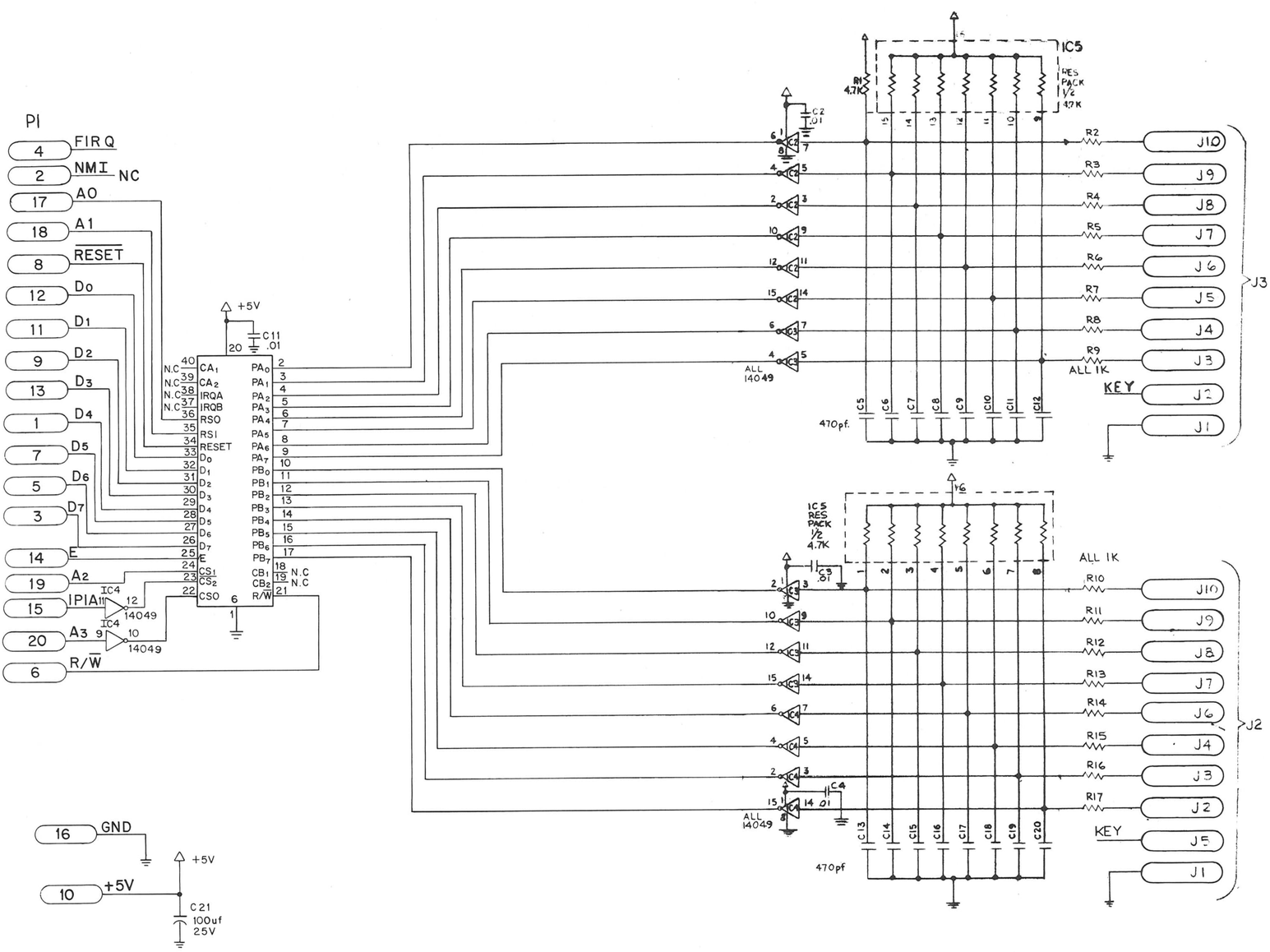


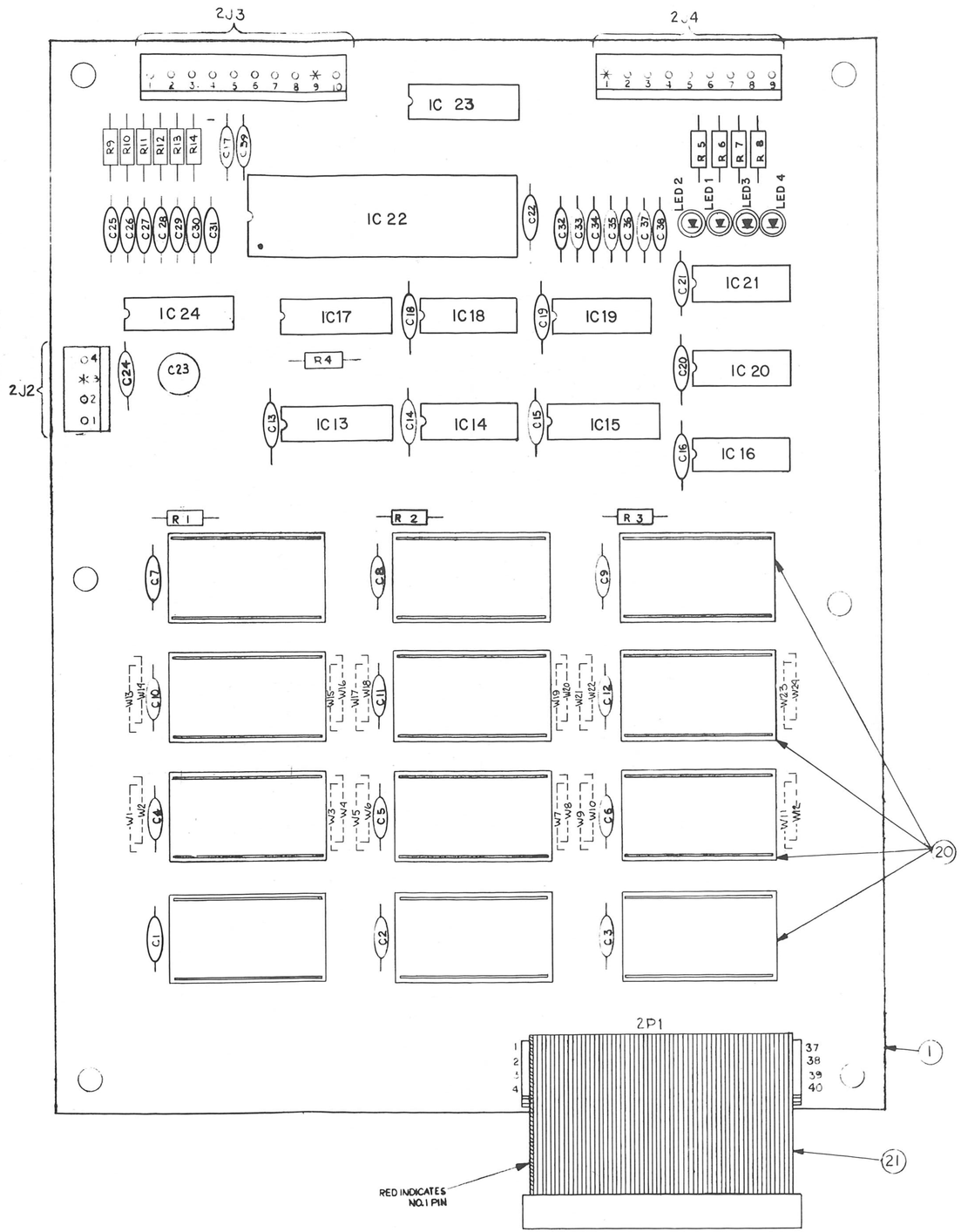
D8356 CPU/Video Board Logic Diagram
(Sheet 2 of 2)

BILL OF MATERIAL

ITEM NO.	PART NO.	PART DESIGNATION	DESCRIPTION	REQ'D NO.
1	5772-09473-00		BARE P.C. BOARD	1
2	5310-08575-00	IC2, IC3, IC4	4049, INV. HEX BUFFER	3
3	5018-09239-00	IC5	4.7K 15 RES, 16 DIP	1
4	5010-08991-00	R1	4.7K Ω RESISTER, ¼ W 5%	1
5	5043-08580-00	C1, C2, C3, C4	.01 MFD. CAPACITOR 50V. AXIAL, CERAMIC +80-20%	4
6	5043-09065-00	C5 THRU C20	470 PFD. CAPACITOR 50V. AXIAL, CERAMIC	16
7	5040-09421-00	C21	100 MFD. CAPACITOR 25V. RADIAL, ELECTROLYTIC	1
8				
9	5791-09444-00	J2 J3	10 PIN HEADER, MOLEX 09-65-1101	2
10	5795-09540-00	3P1	20 PIN RIBBON CABLE ASSY.	1
11	5430-08972-00	IC1	MC6821 PIA	1
12	5010-09358-00	R2 THRU R17	1K OHM RESISTOR ¼ W 5%	16





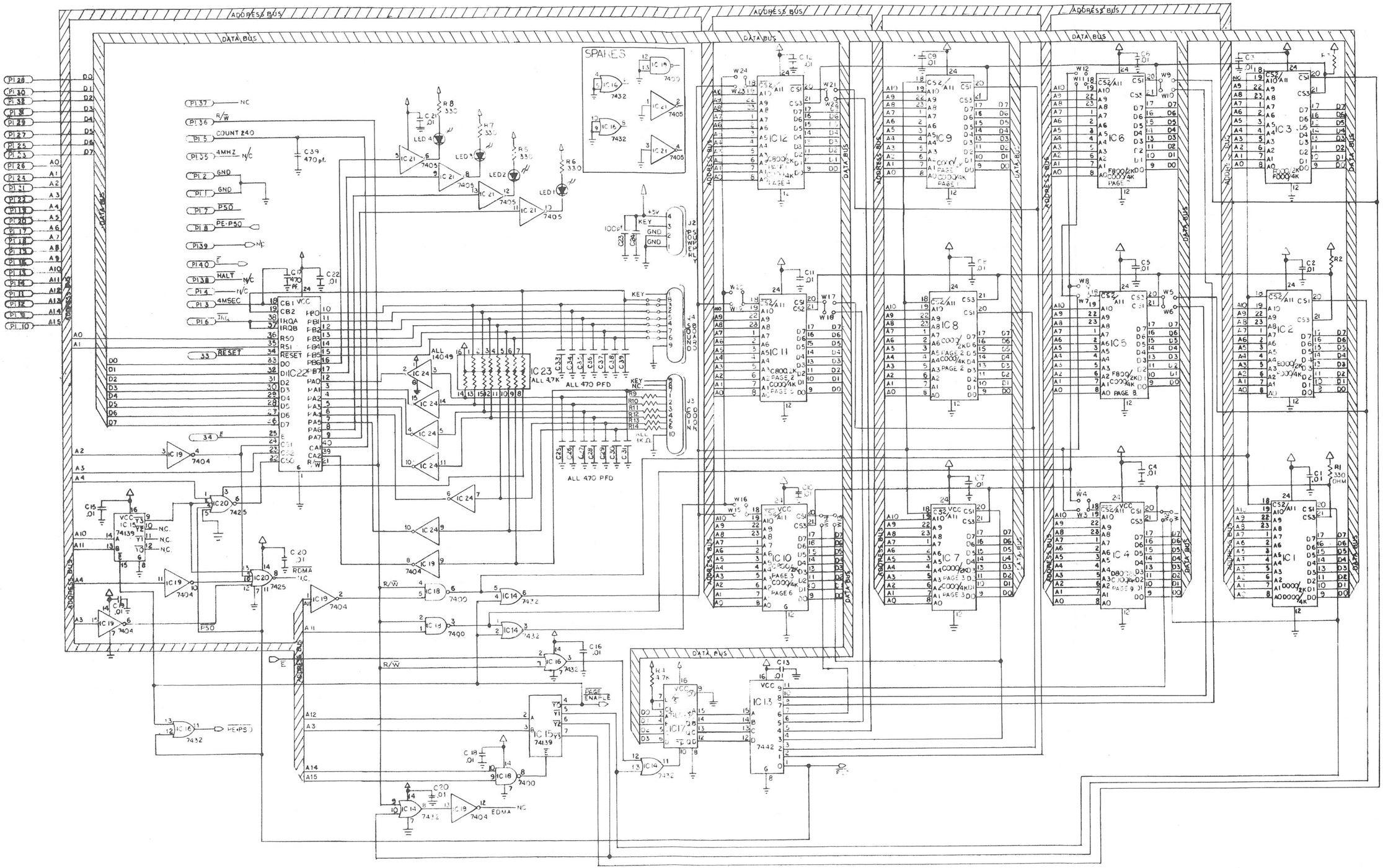


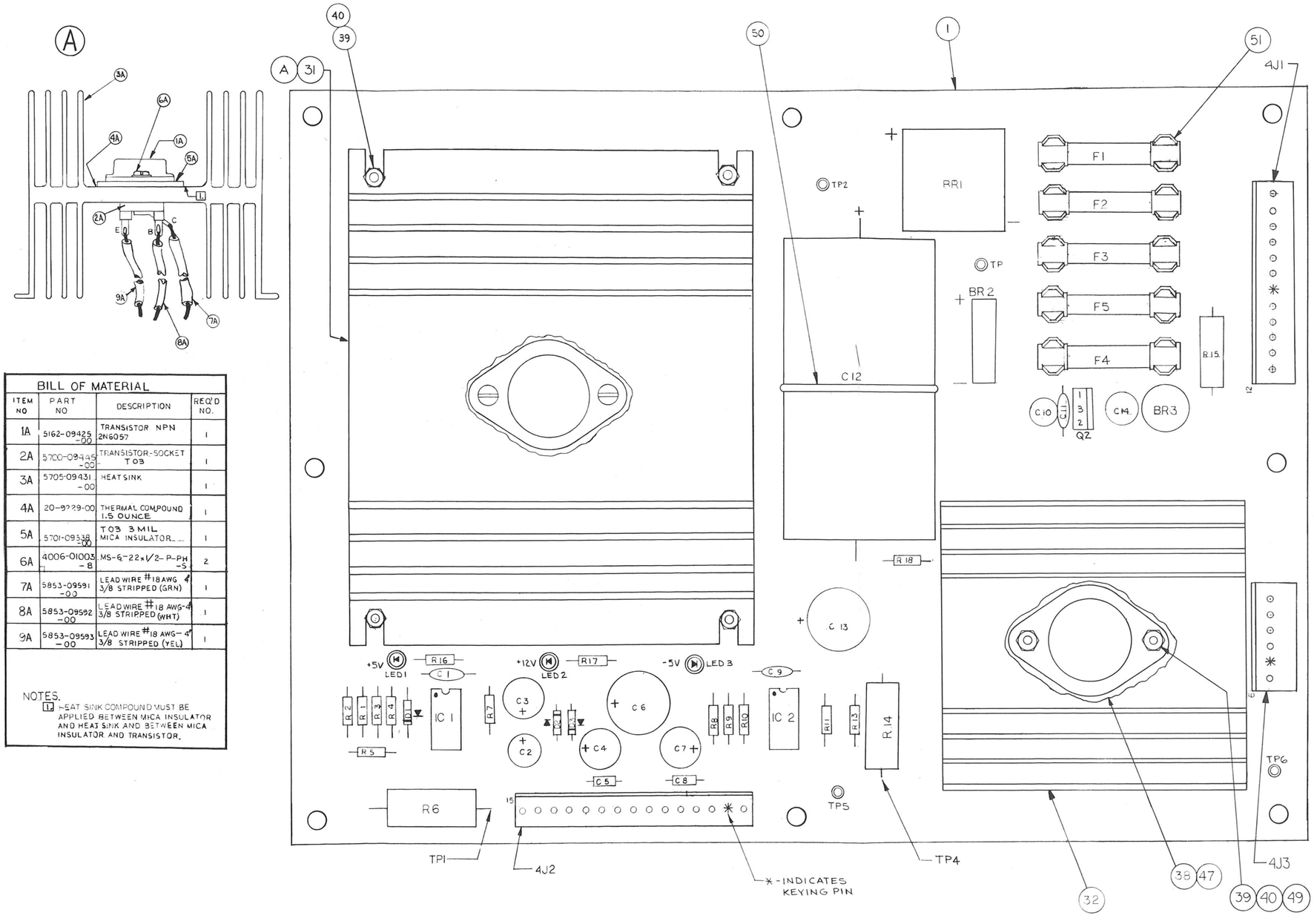
BILL OF MATERIAL

ITEM NO.	PART NO.	PART DESIGNATION	DESCRIPTION	NO. REQ'D
1	5771-09472-00		BARE PC. BOARD	1
2	5280-09012-00	IC 13	7442 BCD DECODER	1
3	5280-08477-00	IC14, IC16	7432 QUADRUPLE 2 INPUT	2
4	5281-09246-00	IC 15	74LS139 DUAL 2 TO 4 LINE DECODER	1
5	5281-09501-00	IC17	74LS395 4-BIT UNIVERSAL SHIFT	1
6	5280-09073-00	IC18	7400 QUAD 2 INPUT NAND GATE	1
7	5280-09013-00	IC 19	7404 HEX, INVERTER	1
8	5280-09551-00	IC 20	7425 NOR GATE WITH STROBE	1
9	5280-08949-00	IC 21	7405 OPEN COLLECTOR OUTPUT	1
10	5430-08972-00	IC 22	MC 6821 PIA	1
11	5019-09239-00	IC 23	16 PIN, 15 RES DIP PACK 4.7K OHM	1
12	5671-09019-00	LED1 THRU LED 4	LED, RED	4
13	5010-09442-00	R1 THRU R3, R5-R8	RESISTOR, C.F. 330 OHM 5% 1/4 W	7
14	5010-08991-00	R4	RESISTOR, C.F., 4.7K OHM 5% 1/4W	1
15				
16	5043-08980-00	C1 THRU C16, C18 THR C22	CAP. CERAMIC .01 MFD. 50V. AXIAL	21
17	5040-09421-00	C 23	CAP. ELECT., 100 MFD. 25V. RAD. +50% -10%	1
18	5043-08996-00	C 24	CAP. CERAMIC, 1 MFD. 50V. AXIAL ±20%	1
19	5043-09065-00	C17, C25 THRU C39	CAP. CERAMIC, 470 PFD. 50V. AXIAL	16
20	5700-09004-00		24 PIN IC SOCKET	12
21	5795-09352-00	2P1	40 PIN RIBBON CABLE ASSEM	1
22	5791-09028-00	2J2	4 PIN MALE CONNECTOR 09-65-1041	1
23	5791-09444-00	2J3	10 PIN MALE CONNECTOR 09-65-1101	1
24	5791-09027-00	2J4	9PIN MALE CONNECTOR 09-65-1091	1
25	5310-08975-00	IC 24	4049 INVERTING HEX. BUFFER	1
26	5010-09358-00	R9 THRU R14	RESISTOR, C.F., 1K OHM 5% 1/4 W.	6
27	SEE CHART	IC1 THRU IC12	SEE CHART	AS REQ'D
28	5010-09534-00	W1 THRU W24	RESISTOR, .00 OHM	AS REQ'D

PART DESIGNATION	GAME NAME	DEFENDER BLUE LABEL	DEFENDER GREEN LABEL			PART NUMBER
	ASSEM NO.	D-8520-00	D-8520-00			
	IC1	5343-09595-00	5343-09621-00			
	IC2	5343-09596-00	5343-09596-00			
	IC3	5343-09597-00	5343-09622-00			
	IC4	NOT USED	5343-09623-00			
	IC5	NOT USED	NOT USED			
	IC6	5343-09600-00	5343-09600-00			
	IC7	5343-09601-00	5343-09601-00			
	IC8	5343-09602-00	5343-09602-00			
	IC9	5343-09603-00	5343-09603-00			
	IC10	5343-09604-00	5343-09604-00			
	IC11	5343-09605-00	5343-09605-00			
	IC12	5343-09606-00	5343-09606-00			
	JUMPERS	W2, W3, W6, W7, W10, W11, W13, W16, W17, W20, W21, W24	W1, W4, W6, W7, W10, W11, W13, W16, W17, W20, W21, W24			

D8520 ROM Board Assembly Drawing


ADDRESS BUS
DATA BUS
SPARES
IC 19 7400
IC 16 7432
IC 21 7405
LED 4
LED 3
LED 2
LED 1
R8 330
R7 330
R5 330
R6 330
C21 .01
C39 470 pf.
PI 37 NC
PI 36 R/W
PI 5 COUNT 240
PI 35 4MHZ N/C
PI 2 GND
PI 1 GND
PI 7 PSO
PI 8 PE-PSO
PI 39 NC
PI 40 E
PI 38 HALT N/C
PI 4 I/C
PI 3 4MSEC
PI 6 IRQ
PI 33 RESET
PI 34 E
PI 32 R/W
PI 31 DMA
PI 30 DMA
PI 29 DMA
PI 28 DMA
PI 27 DMA
PI 25 DMA
PI 23 DMA
PI 26 DMA
PI 24 DMA
PI 21 DMA
PI 22 DMA
PI 19 DMA
PI 20 DMA
PI 17 DMA
PI 18 DMA
PI 15 DMA
PI 16 DMA
PI 13 DMA
PI 14 DMA
PI 11 DMA
PI 12 DMA
PI 9 DMA
PI 10 DMA
C17 470 PF
C22 .01
IC 22 6532
IC 23 ALL 47K
ALL 470 PFD
ALL 14049
IC 24
KEY
J2 POWER
J4 CONN
J3 CONN
R9 R10 R11 R12 R13 R14 ALL 1K Ω
IC 19 7404
IC 20 7425
IC 15 74139
IC 18 7400
IC 14 7432
C15 .01
C20 .01
C18 .01
C16 .01
C14 .01
C19 .01
C23 100pf
C24
+5V
GND
GND
RDMA
PSO
PE-PSO
E
R/W
A11
A12
A3
A14
A15
PAGE ENABLE
IC 17 74LS138
IC 13 7442
R1 7.7K
C13 .01
IC 10 IC 12 IC 9 IC 8 IC 11 IC 6 IC 5 IC 7 IC 4 IC 3 IC 2 IC 1
2KD 4K
W24 W23 W21 W22 W25 W17 W18 W16 W15 W12 W11 W9 W10 W8 W7 W5 W6 W4 W3
C9 C11 C8 C6 C3 C5 C7 C4 C2 C1
R2
R1 330 OHM
PAGE 1
PAGE 2
PAGE 3
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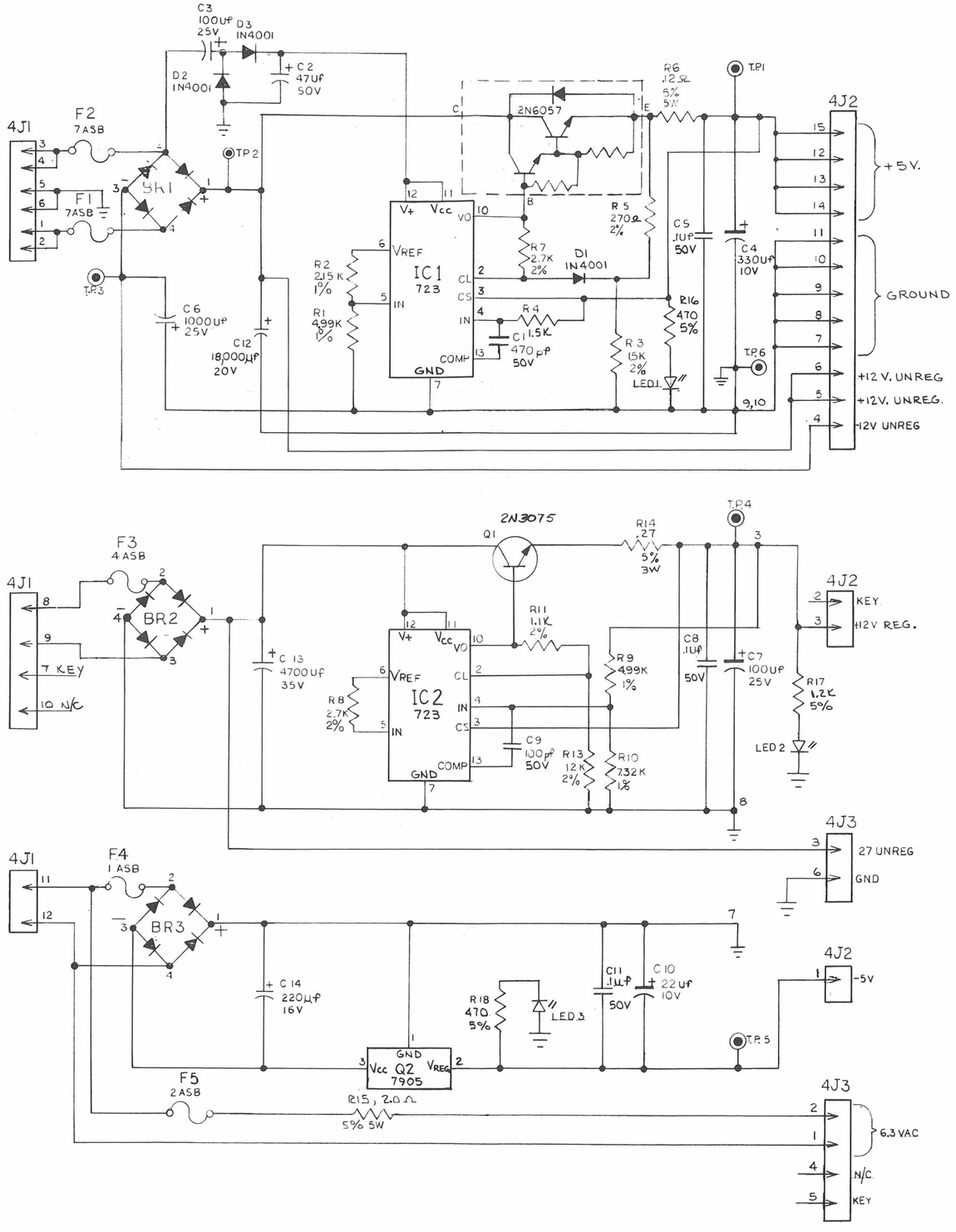
BILL OF MATERIAL

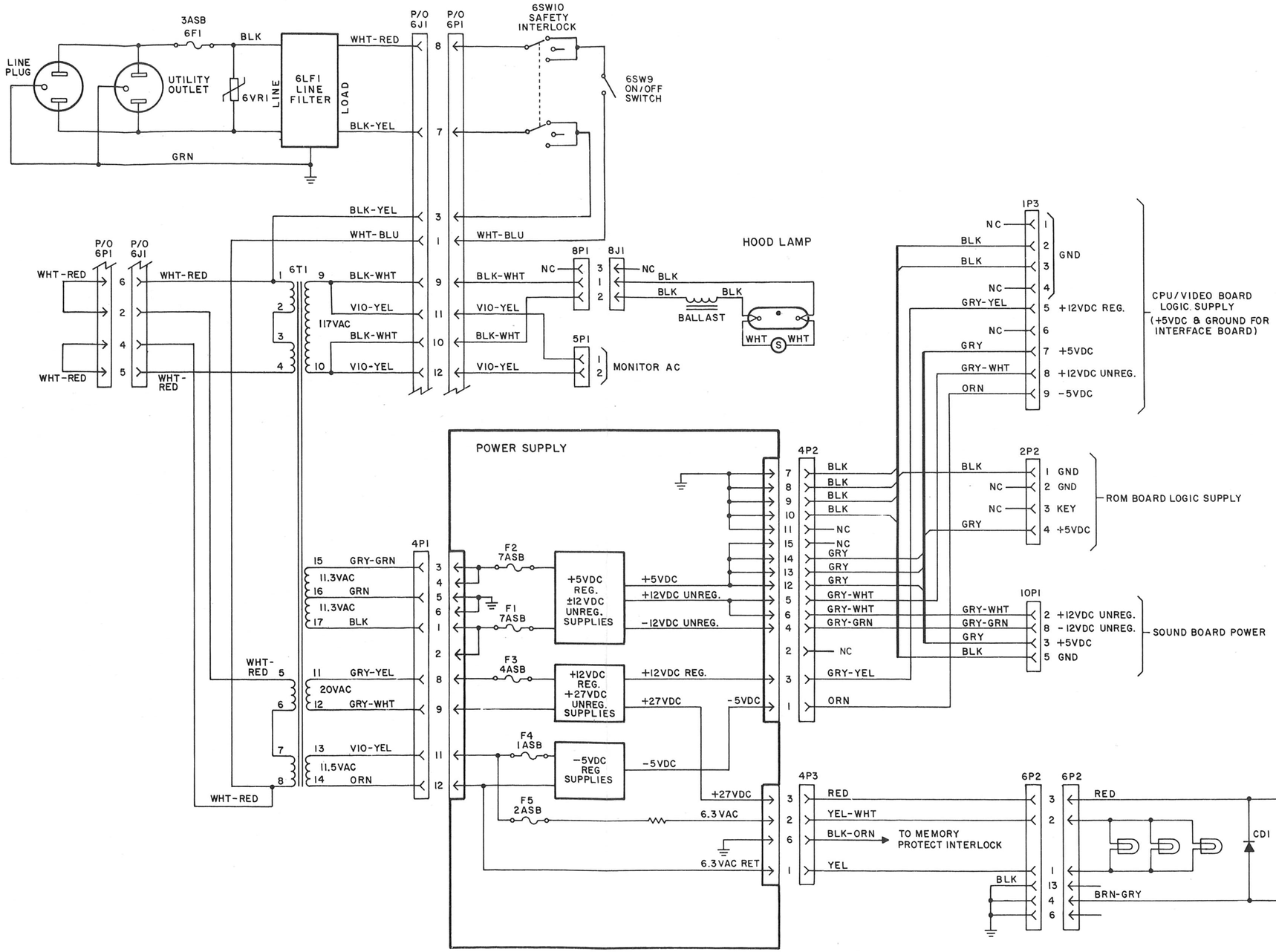
ITEM NO	PART NO	DESCRIPTION	REQ'D NO.
1A	5162-09425-00	TRANSISTOR NPN 2N6057	1
2A	5700-09445-00	TRANSISTOR-SOCKET TO3	1
3A	5705-09431-00	HEAT SINK	1
4A	20-9229-00	THERMAL COMPOUND 1.5 OUNCE	1
5A	5701-09538-00	TO3 3 MIL MICA INSULATOR	1
6A	4006-01003-8	MS-6-22x1/2-P-PH-S	2
7A	5853-09591-00	LEAD WIRE #18AWG 4" 3/8 STRIPPED (GRN)	1
8A	5853-09592-00	LEAD WIRE #18 AWG-4" 3/8 STRIPPED (WHT)	1
9A	5853-09593-00	LEAD WIRE #18 AWG-4" 3/8 STRIPPED (YEL)	1

NOTES.
1. HEAT SINK COMPOUND MUST BE APPLIED BETWEEN MICA INSULATOR AND HEAT SINK AND BETWEEN MICA INSULATOR AND TRANSISTOR.

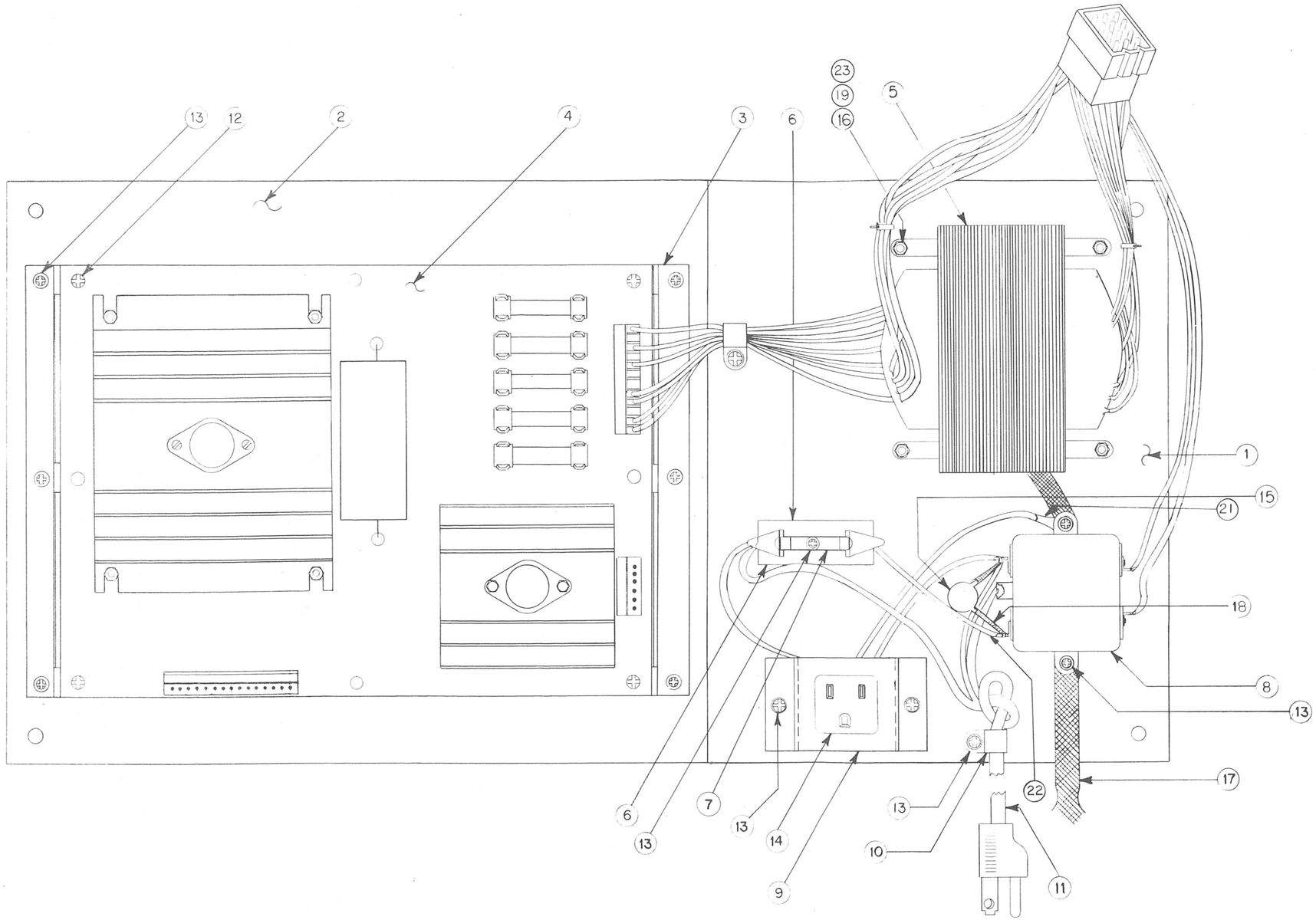
BILL OF MATERIAL

ITEM NO	PART NO.	PART DESIGNATION	DESCRIPTION	REQ'D NO.
1	S773-09474-00		P.S. BARE P.C. BOARD	1
2	5010-09085-00	R4	RESISTOR, 1.5K 5%, 1/4 W	1
3	5010-09541-00	R8, R7	RESISTOR, 2.7K 2%, 1/4W.	2
4	5010-09508-00	R5	RESISTOR, 270 2%, 1/4W	1
5	5010-09428-00	R3	RESISTOR, 1.5K, 2%, 1/4W	1
6	5010-09509-00	R11	RESISTOR, 1.1 K 2%, 1/4W	1
7	5010-09510-00	R13	RESISTOR, 12K 2%, 1/4W.	1
8	5013-09542-00	R10	RESISTOR, 7.32 K 1%, 1/4W	1
9	5013-09427-00	R1 & R9	RESISTOR 4.99K 1% 1/4W	2
10	5013-09426-00	R2	RESISTOR 2.15K 1% 1/4W	1
11	5012-08429-00	R6	RESISTOR 12Ω, 5%, 5W	1
12	5012-09512-00	R14	RESISTOR .27 5%, 5W	1
13	5040-09421-00	C3 & C7	+50% -10% CAPACITOR, 100 MFD, 25V. RADIAL	2
14	5040-09422-00	C2	+50% -10% CAPACITOR, 47 MFD, 50V RADIAL	1
15	5040-09420-00	C6	+50% -10% CAPACITOR, 1000 MFD, 25V. RADIAL	1
16	5040-09419-00	C12	+50% -10% CAPACITOR, 18,000 MFD, 20V. RADIAL	1
17	5040-09423-00	C4	+50% -10% CAPACITOR, 330 MFD, 10V. RADIAL	1
18	5040-09504-00	C13	+50% -10% CAPACITOR, 4700 MFD, 35V. RADIAL	1
19	5043-08994-00	C5, C8, C11	±20% CAPACITOR, .1 MFD, 50V. AXIAL Z5U	3
20	5043-09065-00	C1	±10% CAPACITOR, 470 PFD, 50V. AXIAL X7R	1
21	5043-09492-00	C9	±5% CAPACITOR, 100 PFD 50V. AXIAL S2L	1
22	5040-09506-00	C14	+50% -10% CAPACITOR, 220 MFD, 16V. RADIAL	1
23	5040-09493-00	C10	+50% -10% CAPACITOR, 22 MFD, 10V. RADIAL	1
24	5070-06258-00	D1, D2, D3	DIODE, IN4001	3
25	5100-09418-00	BR1	BRIDGE RECTIFIER 35 AMP 100V.	1
26	5100-09513-00	BR2	DIODE BRIDGE, 4.0A, 50V	1
27	5100-09514-00	BR3	DIODE BRIDGE 1.0A, 50V.	1
28	5731-09432-00	F1, F2	FUSE, 7A SB.	2
29	5731-06314-00	F3	FUSE, 4A, SB., 250 V.	1
30	5731-06569-00	F4	FUSE, 1A, SB	1
31	B-8416-00		HEAT SINK ASSEMBLY (SEE A)	1
32	5705-09543-00		HEAT SINK	1
33	5791-09074-00	4J2	HEADER, 15 PIN, 09-65-1151	1
34	5791-09043-00	4J1	HEADER, 12 PIN, 09-65-1121	1
35	5250-09515-00	Q2	VOLTAGE REGULATOR 7905	1
36	5460-09424-00	IC1, IC2	VOLTAGE REGULATOR 723	2
37	5824-09248-00	TP2, TP3, TP5, TP6	TERMINAL #1502-1 (TEST POST)	4
38	20-9229-00		THERMAL COMPOUND .02 OUNCE	AR
39	4005-01051-07		5-40x7/16 R.H. MECH. SCREW	6
40	4405-01117-00		5-40 HEX NUT	6
41	5161-09552-00	Q1	2N3055 TRANSISTOR NPN	1
42	5731-08665-00	F5	FUSE, 2A. SB	1
43	5791-09038-00	4J3	HEADER, 6 PIN, 09-65-1061	1
44	5012-09544-00	R15	RESISTOR, 2.0Ω, 5%, 5W	1
45	5010-09416-00	R16, R18	RESISTOR, 470Ω, 5%, 1/4W	2
46	5010-09314-00	R17	RESISTOR, 1.2K, 5%, 1/4 W	1
47	5701-09538-00		INSULATOR, MICA, 3 mil	1
48	5671-09019-00	LED1, LED2, LED3	LED, RED	3


C3 100uf 25V
D3 IN4001
D2 IN4001
C2 47uf 50V
4J1
F2 7ASB
F1 7ASB
BR1
TP2
TP3
C6 1000uf 25V
C12 18,000uf 20V
2N6057
R6 .12Ω 5% 5W
TP1
4J2
R5 270Ω 2%
C5 .1uf 50V
C4 330uf 10V
D1 IN4001
R7 2.7K 2%
R2 2.15K 1%
R1 4.99K 1%
IC1 723
R4 1.5K
C1 470pf 50V
R3 15K 2%
R16 470 5%
LED1
TP6
+5V.
GROUND
+12 V. UNREG
+12V. UNREG.
+12V UNREG
2N3075
Q1
R14 27 5% 3W
TP4
F3 4 ASB
BR2
C13 4700uf 35V
R11 1.1K 2%
R9 4.99K 1%
R8 2.7K 2%
IC2 723
C9 100pf 50V
R13 12K 2%
R10 7.32K 1%
C8 .1uf 50V
C7 100uf 25V
R17 1.2K 5%
LED2
7 KEY
10 N/C
KEY
+12V REG.
4J3
27 UNREG
GND
F4 1 ASB
BR3
C14 220µf 16V
R18 470 5%
LED3
C11 .1µf 50V
C10 22uf 10V
-5V
TP5
GND Vcc Q2 VREG 7905
F5 2ASB
R15, 2.0Ω 5% 5W
6.3 VAC
N/C
KEY



Power Wiring Diagram



BILL OF MATERIALS			
ITEM	PART Nº	DESCRIPTION	QTY.
1	11-580	POWER SUPPLY MTG. BD.	1
2	01-7001	POWER SUPPLY GND PLATE	1
3	01-7002	POWER SUPPLY SUPPORT BRKT.	2
4	D. 8359	POWER SUPPLY BOARD	1
5	C 8529	VIDEO TRANSFORMER ASSY	1
6	B-6448-1	FUSE MOUNT ASS'Y.	1
7	5731-08633-00	LINE FUSE 3AMP 115/V	1
8	5102-08895-00	LINE FILTER 5A	1
9	01-6948-	OUTLET MTG. BRACKET	1
10	03-6048-6	CABLE CLAMP-3/8	2
11	SEE NOTE #2	LINE CORD	1
12	4006-01018-04	MS. #6-32 × 1/4 P-RH	4
13	4108-01022-08	SMS #8-32 × 1/2 P-RWH-A	13
14	5851-09184-00	3WIRE SNAP-IN OUTLET	1
15	SEE NOTE #1	VARISTOR	1
16	4408-01117-00	NUT 8-32 HEX	4
17	20-9223-3	6 INCH BRAIDED GROUND WIRE	1
18	20-9256	1 INCH SLEEVING INSULATION	2
19	4703-00008-00	NO#8 EXTERNAL LOCK WASHER	4
21	5825-06523-00	TERMINAL LUG	1
22	5852-09575-00	BLACK-18 AWG JUMPER WIRE-2.5L	1
23	4008-01005-22	MS. 8-32 × 3/8 P-PH	4

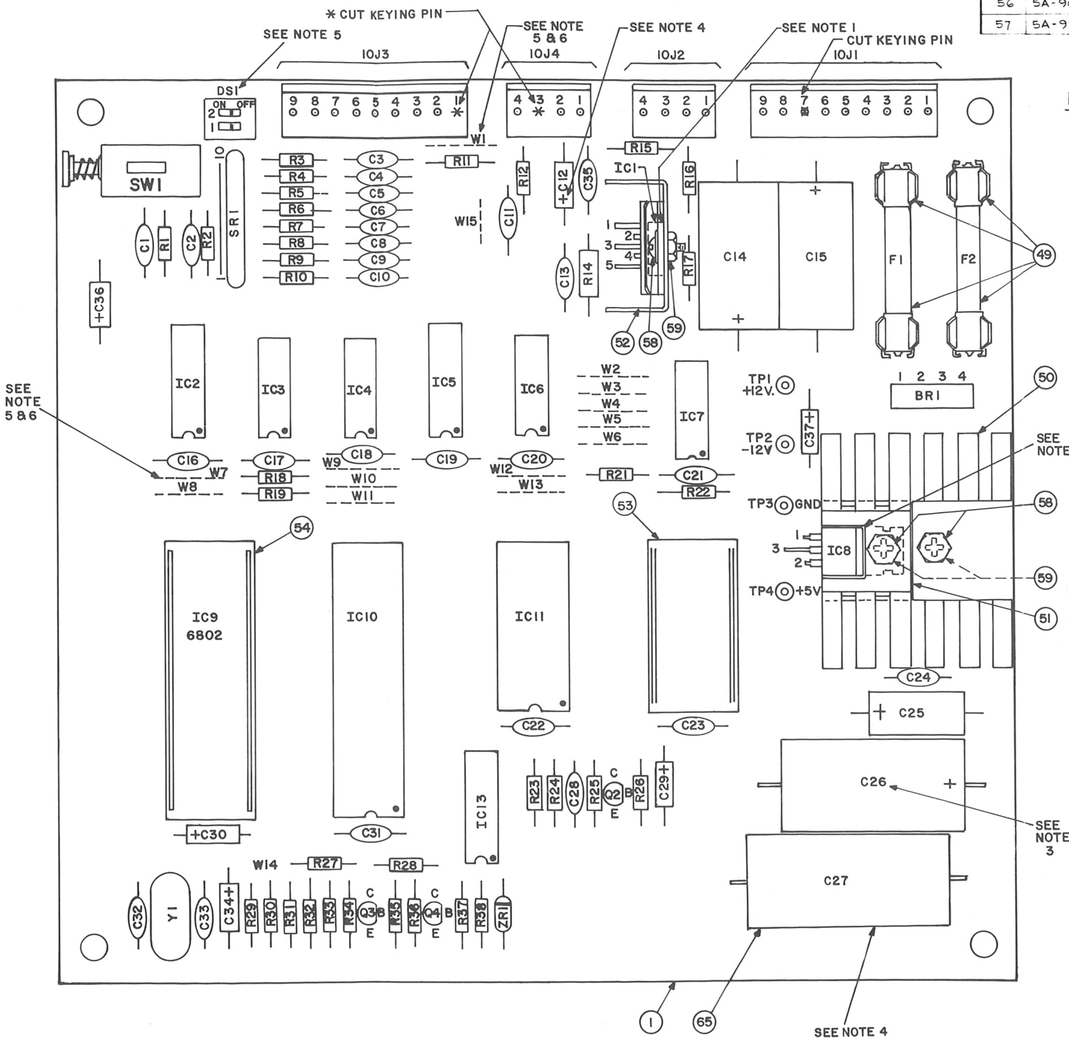
NOTES:

1. FOR 115V.A.C., 130V. VARISTOR # 5017-09044. IS USED.
   FOR 230V.A.C., A 275V VARISTOR # 5017-09063 IS USED.
2. LINE CORD PART NUMBERS:
   - 5850-06206-00 12 FT. EUROPE
   - 5850-08728-00 U.S. DOM.
   - 5850-08737-00 AUS.
   - 5850-08743-00 U.S. L.A.
   - 5850-09128-00 12 FT. FRANCE
   - 5850-09131-00 AWG 18-12 JAPAN

D8373 Power Panel Assembly Drawing

ITEM NO.	PART NO.	PART DESIGNATION	DESCRIPTION	REQ'D. NO.
58			6-32 x 3/16" BINDER HEAD SCREW	3
59			6-32 HEX NUT	3
60		W1, W2, W5, W7, W9, W10, W15	WIRE JUMPER 22 GAUGE WIRE WITH INSULATION	7
61	5A-9248	TP1 THR TP4	TERMINAL # 1502-1	4
62	5A-9363	R11	RESISTOR, FC, 5.6 K OHM 5% 1/4 WATT	1
64	5A-9362	SR1	RESISTOR, 4.7 K OHM 10 PIN SIP	1
65	3A-7520-1		TIE WRAP	1

ITEM NO.	PART NO.	PART DESIGNATION	DESCRIPTION	REQ'D. NO.
48	5A 6314	F1, F2	4 AMP SLOW BLOW FUSE	2
49	5A-9178		FUSE HOLDER	4
50	5A-9172		HEAT SINK THERMALLOY #6072B	1
51	5A-9173		HEAT SINK THERMALLOY #6071B	1
52	5A-9199		HEAT SINK THERMALLOY #6030	1
53	5A-9004		24 PIN SOCKET	1
54	5A-8985		40 PIN SOCKET	1
55	5A-9027	10J1, 10J3	9 PIN MALE CONNECTOR	2
56	5A-9028	10J2, 10J4	4 PIN MALE CONNECTOR	2
57	5A-9349	10J5	40 PIN RIBBON HEADER	1



NOTES:

1. USE THERMAL COMPOUND BETWEEN IC'S AND HEAT SINK.
2. CAUTION: AVOID STATIC DISCHARGE DAMAGE TO MOS LOGIC.
3. SYMBOLS SHOWN ON COMPONENTS ARE FOR REFERENCE ONLY. DO NOT SCREEN OR STAMP.
4. OBSERV. INDEX MARK OF ALL INTEGRATED CIRCUITS, DIODES D1, D2, AND ZR1.

   CAPACITORS C12, C13, C15, C25, C29, C27.

   CONNECTORS 10J1, 10J2, 10J4, 10J3, 10J5.

   POSITION OF TRANSISTORS Q1, Q2, Q3, Q4.
5. DS1 - 1 SELECTS SOUNDS/NOTES
   2 SELECTS SPEECH/NO SPEECH (W9/W11)

   W7 - SPEECH MODULE STATUS
   IN - SPEECH MODULE NOT ATTACHED
   OUT- SPEECH MODULE ATTACHED

   W14 - MPU INTERNAL RAM ENABLE

   W7 & W8-MEMORY MAP CONTROL

   W12 & W13 - PB7 STATUS CONTROL (W13 NEVER USED)

   W4 & W9 - PB5 STATUS CONTROL
6. SOLDERED ON TOP OF BOARD

INSTAL THESE JUMPERS FOR FOLLOWING GAMES:

W15, W8, W12, W4, W1, W3, W6, W11 FOR:

WORLD CUP DISCO FEVER CONTACT POKERINO PHOENIX ARISTOCRAT SHUFFLE POMPELI SHUFFLE KING TUT SHUFFLE TAURUS SHUFFLE	SOUND ROM 1

W15, W8, W12, W4, W1, W2, W5, W10, FOR:

FLASH STELLAR WARS TRI ZONE TIME WARP	SOUND ROM 1 SA-9198

W7, W15, W8, W1, (SEE NOTE #5) W2, W5, W10 FOR:

GORGAR; SOUND ROM 2 - SA-9360

IC #2 SELECTION STRAPPING:

(2K x 8)	(4K x 8)	(512 x 8)
W2	W3	W3
W7 IN	W8 IN	W8 IN
W1	W5	W10

REVISION LETTER	REVISION	BY
E	REVISED NOTES, ADDED ITEM #45 & NOTE #6, RELOCATED W15, IN W13, DASH LINE WAS SOLID LINE & ITEM #60, W15 WAS W13 E.C.O. 11-19-79	R.GAY
D	DELETED ITEM #14 PT. NO. 5C-9016; ITEM #16 PT. NO. 5A-8910; ITEM #25, PT. NO. 5A-9344 ITEM #45 PT. NO. 5A-9360 & IN ITEM #21, DELETED R13 QTY. WAS 6 E.C.O. 9764 11-14-79	R.GAY
C	ADDED C36, ITEM #34 PT NO WAS 5A-9031 & QTY. WAS 1, & C30 WAS .01 MFD. E.C.O. 9760 10-29-79	K.GAY
B	DELETED ITEM #62 ADDED R13 & R12 TO ITEM #14 QTY. WAS 5 10-2-79	K.G.
A	ITEM #1, P.C. BD. NO. WAS 5A-9151 ITEM #32, ADDED C12 ITEM #60 HOOKUP WIRE 10-2-79	K.G.

DIETZGEN NO. 188H AGEPROOF

BILL OF MATERIAL

ITEM NO.	PART NO.	PART DESIGNATION	DESCRIPTION	REQ'D. NO.
1	IC-2001-146-3		BARE P.C. BOARD	1
2	5A-9156	IC1	TDA2002 V AUDIO AMPLIFIER	1
3	5A-9012	IC2	7442 BCD-DEC DECODER	1
4	5A-9073	IC3	7400 QUAD 2 INPUT NAND	1
5	5A-8973	IC4	7408 QUAD 2 INP. AND GATE	1
6	5A-9153	IC5	4050 BUFFER	1
7	5A-9154	IC6	4068 8 INPUT NAND GATE	1
8	5A-8971	IC7	14069 HEX INVERTER	1
9	5A-9157	IC8	7805 5 VOLT REG. W/TO220 CASE	1
10	5A-8972	IC10	6821 P.I.A.	1
11	5A-9003	IC11	6810 RAM	1
12	5A-9152	IC13	1408 D/A CONVERTER	1
13	5C-8936	Q2, Q3, Q4	2N4401 NPN TRANSISTOR	3
14				
15	5A-9018	ZR1	IN5996 6.8V. ZENER DIODE	1
16				
17	5A-9158 OR 5A-9357	BR1	MDA 200/3N253 BRIDGE RECTIFIER	1
18	5A-9020	Y1	3.58 MHz CRYSTAL	1
19	5B-8991	R1, R18, R19, R21, R22, R27, R32, R33, R34	RESISTOR, FC, 4.7K OHM 5% 1/4 WATT	9
20	5B-9036	R2 THRU R10	RESISTOR, FC, 100 OHM 10% 1/4 WATT	9
21	5A-8984	R12, , R15, R28, R36, R38	RESISTOR, FC, 1K OHM 10% 1/4 WATT	5
22	5A-9181	R14	RESISTOR, FC, 1 OHM 10% 1/2 WATT	1
23	5A-9161	R16	RESISTOR, FC, 2.2 OHM 10% 1/4 WATT	1
24	5A-9361	R17	RESISTOR, FC, 220 OHM 10% 1/2 WATT	1
25				
26	5B-8983	R23, R24, R26	RESISTOR, FC, 3.3K OHM 10% 1/4 WATT	3
27	5A-9179	R25	RESISTOR, FC, 3.3M OHM 10% 1/4 WATT	1
28	5A-9359	R29	RESISTOR, FC, 47 K OHM 5% 1/4 WATT	1
29	5B-8817	R33, R35, R37	RESISTOR, FC, 10 K OHM 10% 1/4 WATT	3
30	5B-9039	R34	RESISTOR, FC, 10 OHM 10% 1/4 WATT	1
31	5A-8980	C1, C16 THRU C28 C31	CAPACITOR, CERAMIC, .01 MFD. 50 V. ±20%	11
32	5A-9065	C2 THRU C10	CAPACITOR, CERAMIC, 470 PFD. 50 V. ±20%	9
33	5A-9345	C11	CAPACITOR, CERAMIC, .001 MFD. 20% 100 V.	1
34	5A-9365	C12, C30, C36	CAPACITOR, ELECTROLYTIC 1 MFD. 63 V. -10/+ 50%	3
35	5A-8996	C13, C24, C35	CAPACITOR, CERAMIC, .1 MFD. 50 V. ±20%	3
36	5A-9165 5A-9165-1	C14	CAPACITOR, ELECTROLYTIC, 800 MFD. 16 V. OR 1,000 MFD. 15 V. ±20%	1
37	5A-9164 5A-9164-1	C15	CAPACITOR, ELECTROLYTIC, 500 MFD. 15 V. OR 470 MFD. 25 V. ±20%	1
38	5A-8986	C25	CAPACITOR, ELECTROLYTIC, 100 MFD. 10 V. ±20%	1
39	5A-8893	C26,	CAPACITOR, ELECTROLYTIC, 1,000 MFD. 25 V. ±20%	1
40	5A-9046	C27	CAPACITOR, ELECTROLYTIC, 12,000 MFD. 16 V. ±20%	1
41	5A-9180	C28	CAPACITOR, CERAMIC, 47 PFD. 1K V. ±20%	1
42	5A-9343	C29	CAPACITOR, ELECTROLYTIC, 10 MFD. 25 V. LOW LEAK. ±20%	1
43	5A-9169	C32, C33	CAPACITOR, CERAMIC DISC, 27 PFD. 1K V. ±10%	2
44	5A-9163	C34	CAPACITOR, TANTALUM, 2.2 MFD. 15 V. ±20%	1
45	5A-9031	C37	CAPACITOR, TANTALUM, 1 MFD. 25 V. ±20%	1
46	5A-9024	SW1	MOMENTARY SWITCH SPDT	1
47	5A-9330	DS1	2 STD. DIP SWITCH	1

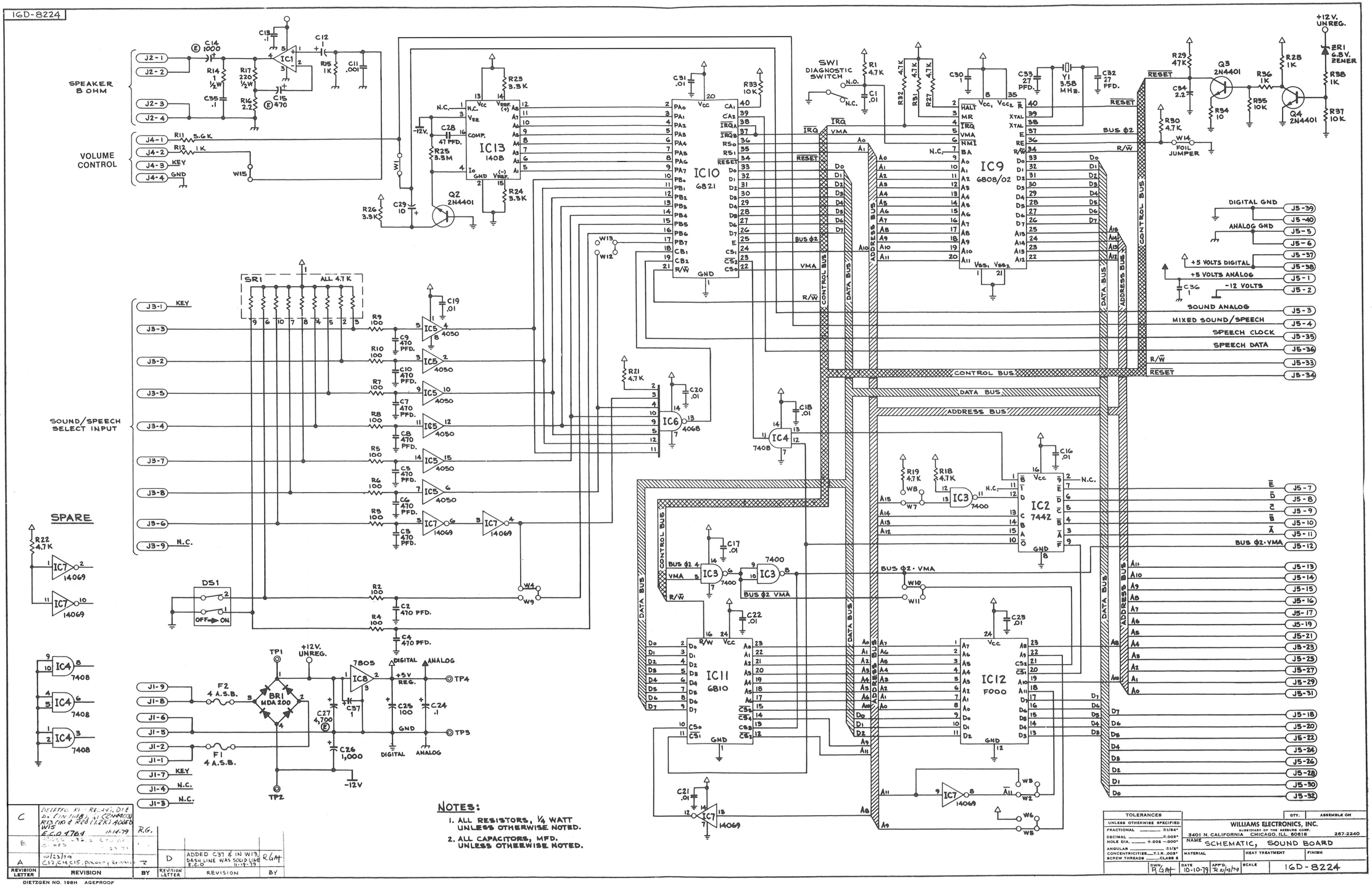
TOLERANCES	QTY.	ASSEMBLE ON
UNLESS OTHERWISE SPECIFIED FRACTIONAL ±1/64" DECIMAL .XX ±.005" HOLE DIA. .XXX +.003 -.000" ANGULAR ±1/2° CONCENTRICITIES .T.I.R. .003" SCREW THREADS CLASS 2		

WILLIAMS ELECTRONICS, INC.
SUBSIDIARY OF THE SEEBURG CORP.
3401 N. CALIFORNIA CHICAGO, ILL. 60618 267-2240

NAME SOUND BOARD SUB-ASSEM

MATERIAL | HEAT TREATMENT | FINISH

DWN. RGA | DATE 9-23-79 | APP'D. | SCALE 2:1 | D-8223



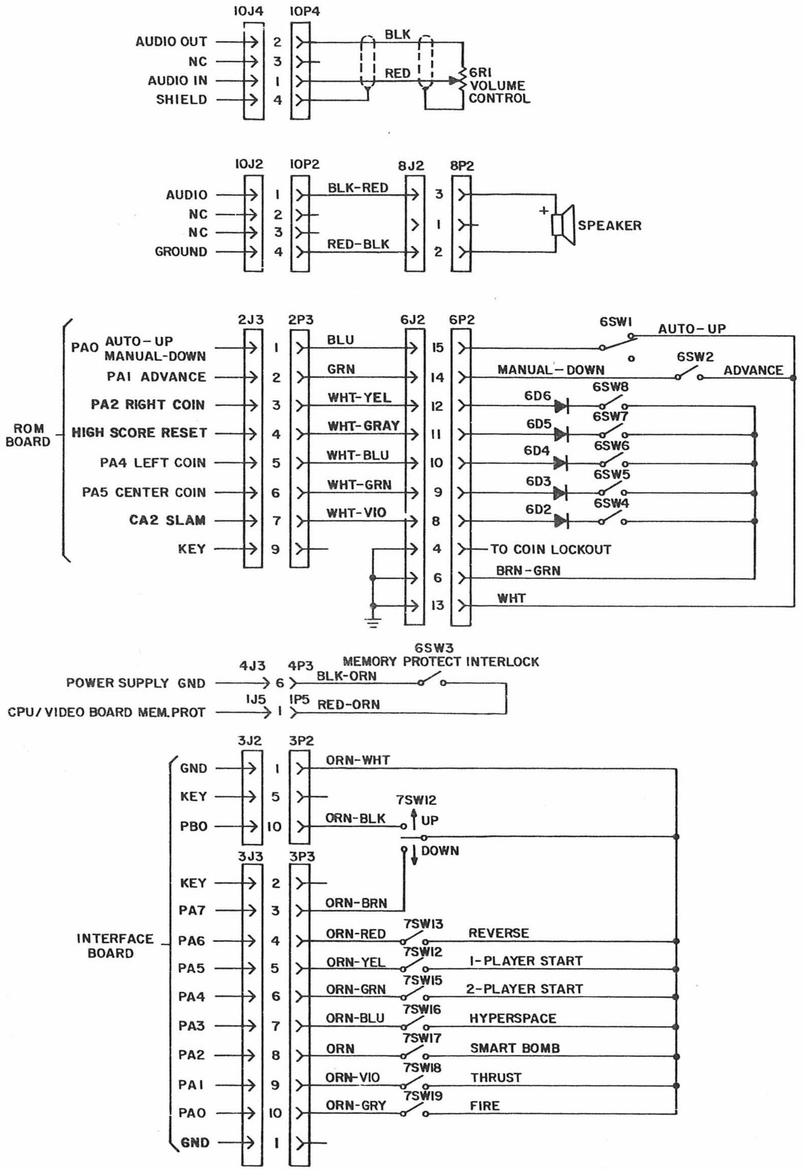
NOTES:

1. ALL RESISTORS, ¼ WATT UNLESS OTHERWISE NOTED.
2. ALL CAPACITORS, MFD. UNLESS OTHEERWISE NOTED.

REVISION LETTER	REVISION	BY
C	DIFFERENTIAL RELAY, DIE D: C(IN 1118), R3 (CHANGED WITH R23 & R26 (1.2K) ADDED E.C.O 4764 11-14-79	R.G.
B	ADDED C37 & IN W13, DASH LINE WAS SOLID LINE E.C.O 11-19-79	R.G.M.
A	10/23/79 C12, C14, C15 PREVIOUSLY	

DIETZGEN NO. 198H AGEPROOF

TOLERANCES UNLESS OTHERWISE SPECIFIED		QTY.	ASSEMBLE ON
FRACTIONAL ±1/64"	WILLIAMS ELECTRONICS, INC. SUBSIDIARY OF THE BEBURNE CORP. 3401 N. CALIFORNIA CHICAGO, ILL. 60618 267-2240		
DECIMAL ±.005"	NAME SCHEMATIC, SOUND BOARD		
HOLE DIA. +.003 −.000			
ANGULAR ±1/2°	MATERIAL	HEAT TREATMENT	FINISH
CONCENTRICITIES T.I.R. .005"			
SCREW THREADS CLASS 2			
DWN. R.G.M.	DATE 10-10-79	APP'D. 10/19/79	SCALE
			16D-8224



Cabinet Wiring Diagram